

M: Byte Flu

Source file name: `flu.c`, `flu.cpp`, `flu.java`, or `flu.py`

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A computer virus known as Byte Flu has discovered a vulnerability in the identification system used by a company's network. The network consists of N devices, numbered from 1 to N , each with a unique string identifier. Byte Flu can move directly between two different devices if and only if the identifier of one can be transformed into the identifier of the other using at most K edit operations. Each edit operation consists of inserting one letter at any position, deleting one letter, or replacing one letter with a different one. Intermediate strings produced during a transformation do not need to be identifiers of network devices.

Some devices are *critical* because they provide services that are essential to the network. Consequently, critical devices cannot be disconnected.

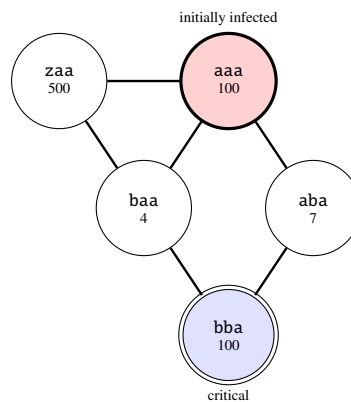
Initially, some non-critical devices are infected with Byte Flu. Whenever the virus reaches a device, it can move directly from that device to any other device whose identifier satisfies the condition above. This process continues, so the virus can eventually reach a device through a sequence of direct movements involving other devices. If a device is disconnected, the virus can neither enter it nor use it to reach another device. Initially infected devices cannot be disconnected.

Each device i has an associated disconnection cost c_i . To protect the network, the managers may disconnect any collection of devices that are neither critical nor initially infected. The cost of doing so is the sum of the disconnection costs of all disconnected devices. Their goal is to ensure that Byte Flu cannot reach any critical device from any initially infected device while minimizing the total disconnection cost. In some situations, protecting all critical devices may be impossible.

For example, consider a network with five devices whose identifiers and disconnection costs are

(aaa, 100), (baa, 4), (aba, 7), (bba, 100), (zaa, 500).

For aaa initially infected, bba critical, and $K = 1$, the network is depicted next:



Byte Flu can reach bba through the sequences

aaa $\rightarrow$ baa $\rightarrow$ bba

aaa $\rightarrow$ aba $\rightarrow$ bba.

If only baa is disconnected, the virus can still use the second sequence; if only aba is disconnected, it can still use the first. Disconnecting both devices prevents Byte Flu from reaching bba, at a total cost of $4 + 7 = 11$. No valid solution has a lower cost.

Given the device identifiers and disconnection costs, the initially infected devices, the critical devices, and the reachability threshold K , can you write a program that determines the minimum total cost required to prevent Byte Flu from reaching any critical device? If this is impossible, your solution must also report it.

Input

The input consists of several test cases. Each test case begins with a line containing four integers N ($2 \leq N \leq 1000$), K ($0 \leq K \leq 20$), I ($1 \leq I$), and C ($1 \leq C$ and $I + C \leq N$) representing the number N of devices, the reachability threshold K , the number I of initially infected devices, and the number C of critical devices.

The next N lines describe the devices. The i -th of these lines contains a string s_i ($1 \leq |s_i| \leq 20$) of lowercase English letters and an integer c_i ($1 \leq c_i \leq 10^6$), representing the identifier and the disconnection cost of device i , respectively. All device identifiers in a test case are distinct.

The next line contains I pairwise distinct integers $p_1, p_2, \dots, p_I$ ($1 \leq p_j \leq N$) identifying the initially infected devices. The following line contains C pairwise distinct integers $q_1, q_2, \dots, q_C$ ($1 \leq q_j \leq N$), identifying the critical devices. No device is both initially infected and critical. Although a disconnection cost is provided for every device, initially infected and critical devices cannot be disconnected.

The end of the input is indicated by a line with four blank-separated zeros and should not be processed.

The input must be read from standard input.

Output

For each test case, output one line containing a single integer: the minimum total cost of disconnecting devices so that Byte Flu cannot reach any critical device from any initially infected device. If no device needs to be disconnected, output 0. If it is impossible to prevent the virus from reaching all critical devices, output -1 .

The output must be written to standard output.

Sample Input	Sample Output
5 1 1 1 aaa 100 baa 4 aba 7 zaa 500 bba 100 1 5 4 2 1 1 aaaa 100 aabb 6 bbbb 100 zzzz 10 1 3 0 0 0 0	11 6